

LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO: 34. Morphological Operation Using Top Hat

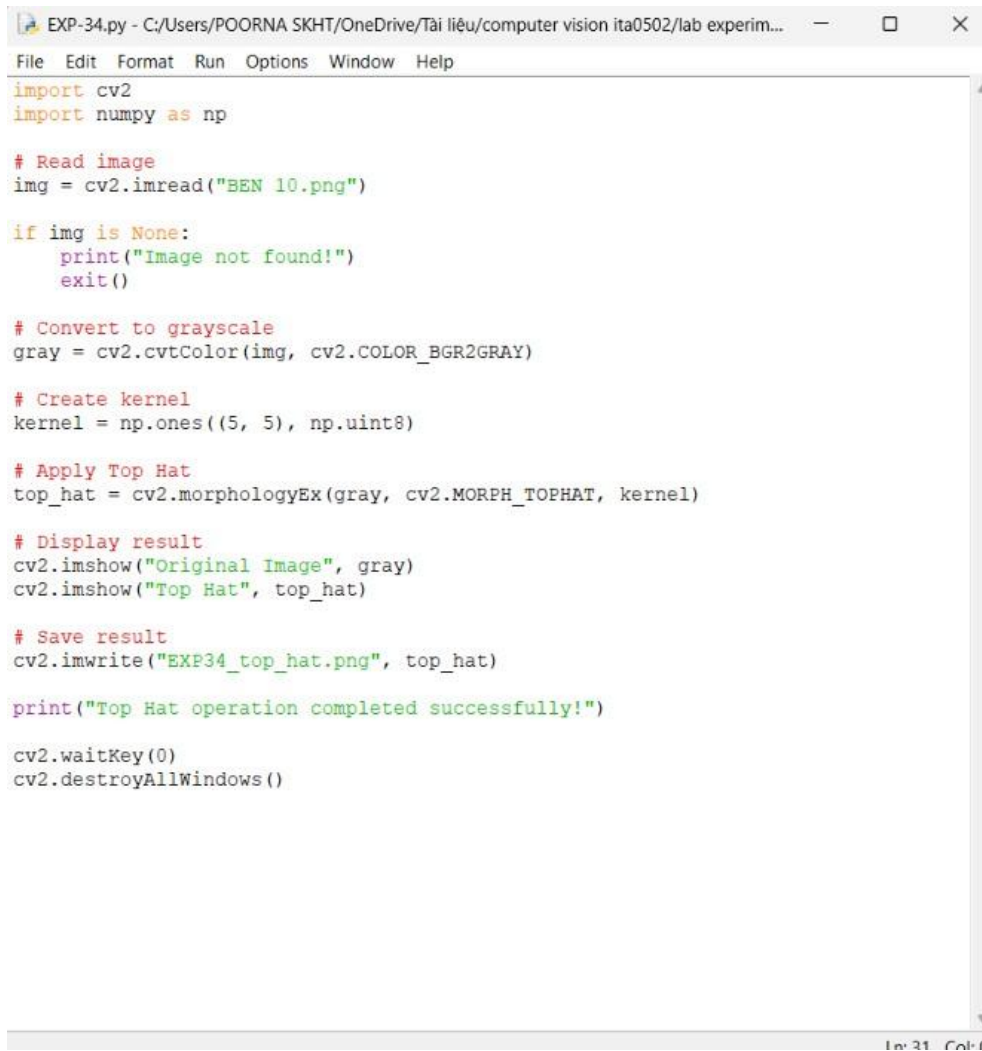
Aim

To perform Top Hat morphological operation on an image using OpenCV.

Algorithm

1. Read BEN 10.png.
2. Convert the image to grayscale.
3. Create a morphological kernel.
4. Apply Top Hat operation.
5. Display and save the result.

CODE:



```
EXP-34.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...
File Edit Format Run Options Window Help
import cv2
import numpy as np

# Read image
img = cv2.imread("BEN 10.png")

if img is None:
    print("Image not found!")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Apply Top Hat
top_hat = cv2.morphologyEx(gray, cv2.MORPH_TOPHAT, kernel)

# Display result
cv2.imshow("Original Image", gray)
cv2.imshow("Top Hat", top_hat)

# Save result
cv2.imwrite("EXP34_top_hat.png", top_hat)

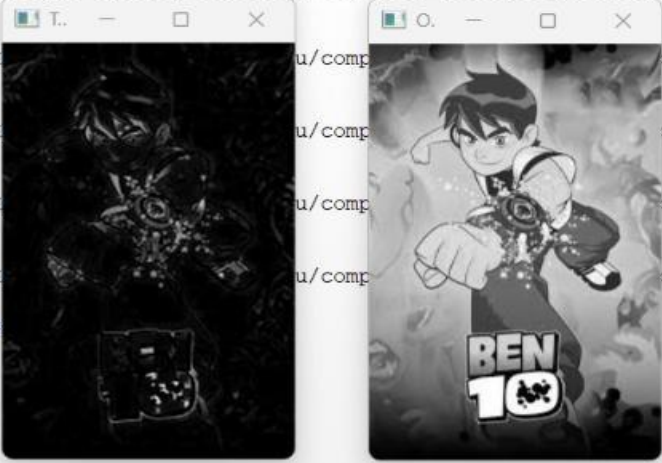
print("Top Hat operation completed successfully!")

cv2.waitKey(0)
cv2.destroyAllWindows()
```

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OUTPUT :

```
*IDLE Shell 3.13.3*
File Edit Shell Debug Options Window Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-26.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-28.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-29.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-30.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-31.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-32.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-33.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-34.py
Top Hat operation complete
```



Result

The Top Hat morphological operation is successfully performed to highlight small bright regions in the image.